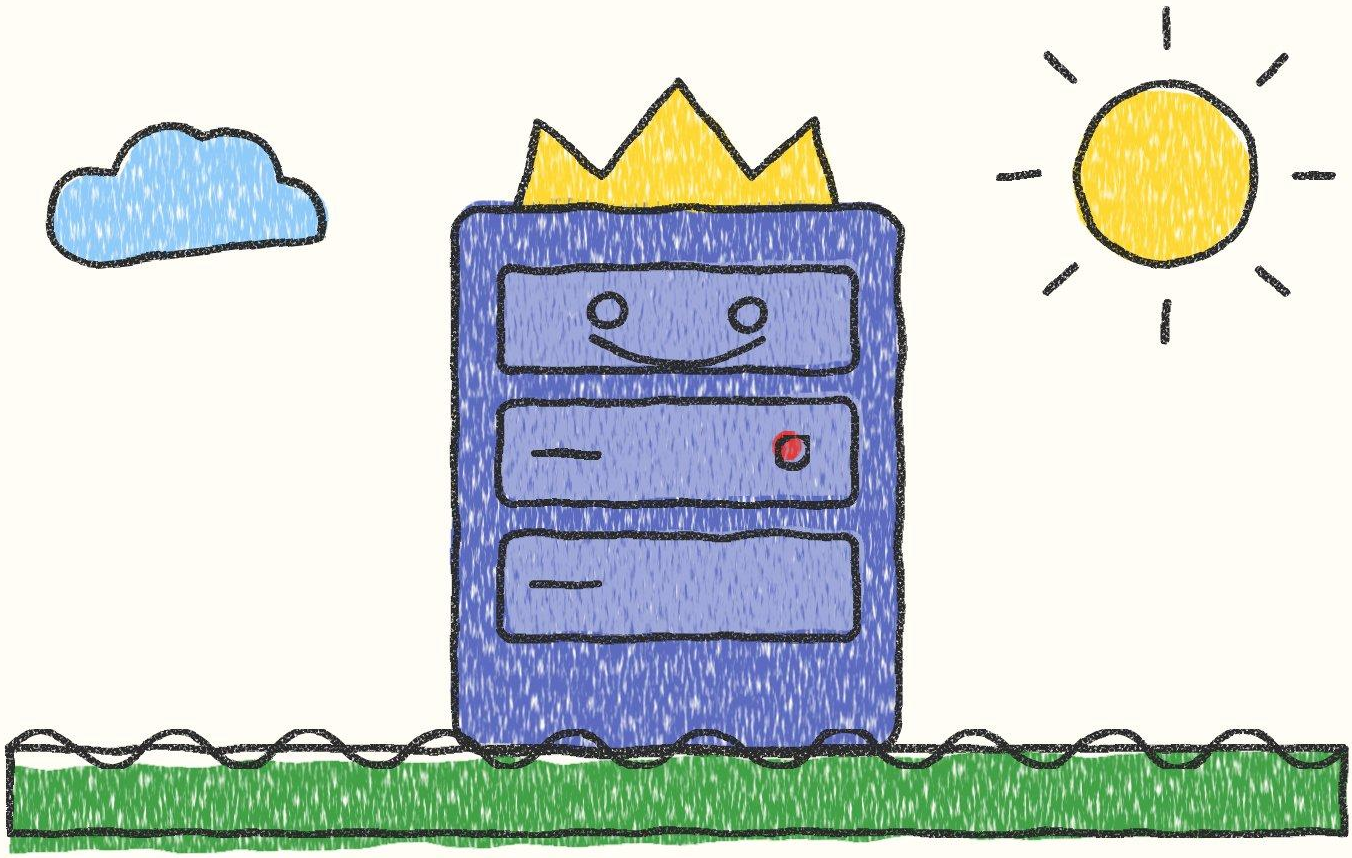
The Big Book of MCP Server Portals

A coloring book about one front door for all your AI tools



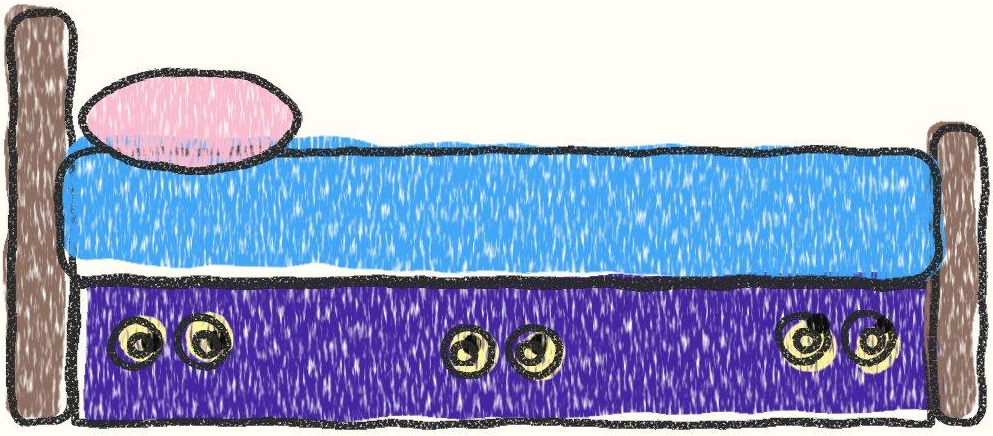
Based on real, public Cloudflare docs. One URL, lots of servers, zero fire trucks. Please stay inside the lines. (We will not.)

Color me: everything. Budget: unlimited crayons.

Adapted from Cloudflare docs (CC BY 4.0). Ages 18+ and SREs.

Shadows Under The Bed

EEK!

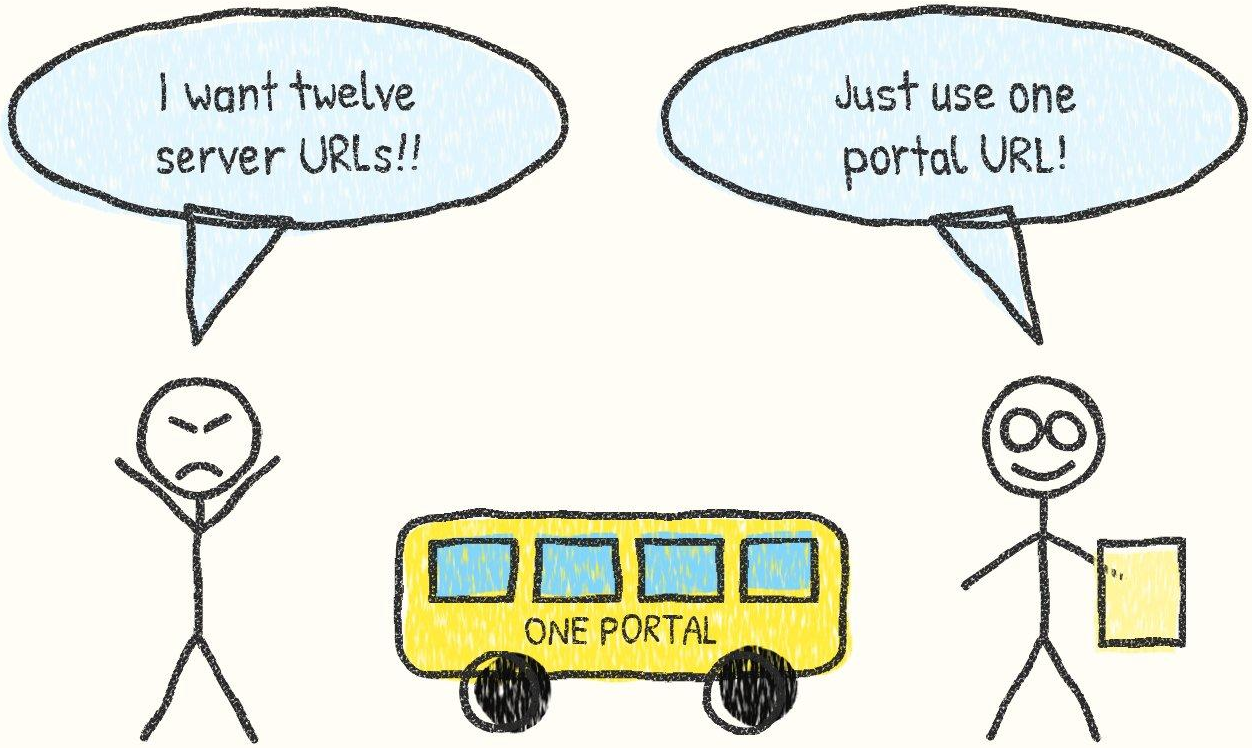


Every team plugs its AI helper into lots of MCP servers, one at a time, and nobody has the whole list. Some servers hide sneaky instructions in their tool descriptions. Some are left wide open on the network. EEK.

Color me: spooky purple under the bed

Grown-up version: Cloudflare blog, "Securing the AI Revolution"

One Bus, Not Twelve Ponies

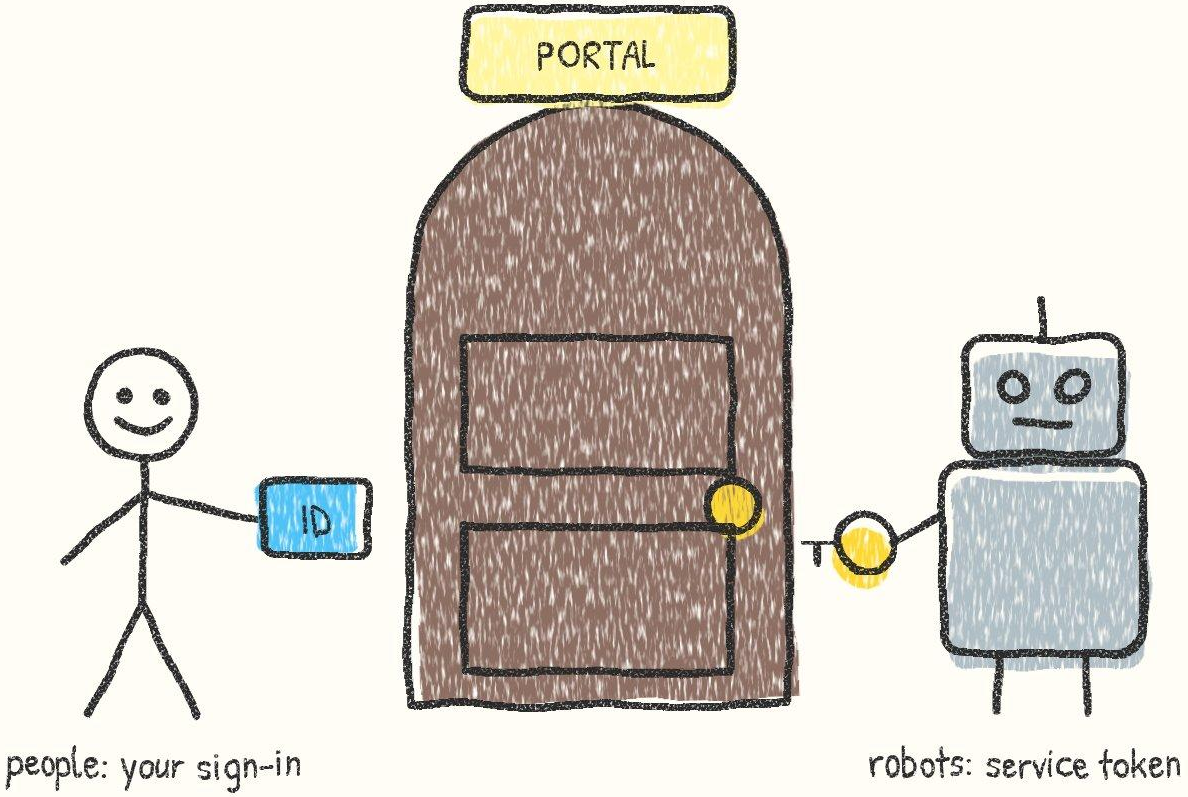


Admins put the approved MCP servers into one portal. You point your MCP client at one address that ends in /mcp, and ride the bus to all of them.

Color me: bus yellow, obviously

Grown-up version: Cloudflare MCP server portals docs, overview

Knock Once At The Front Door

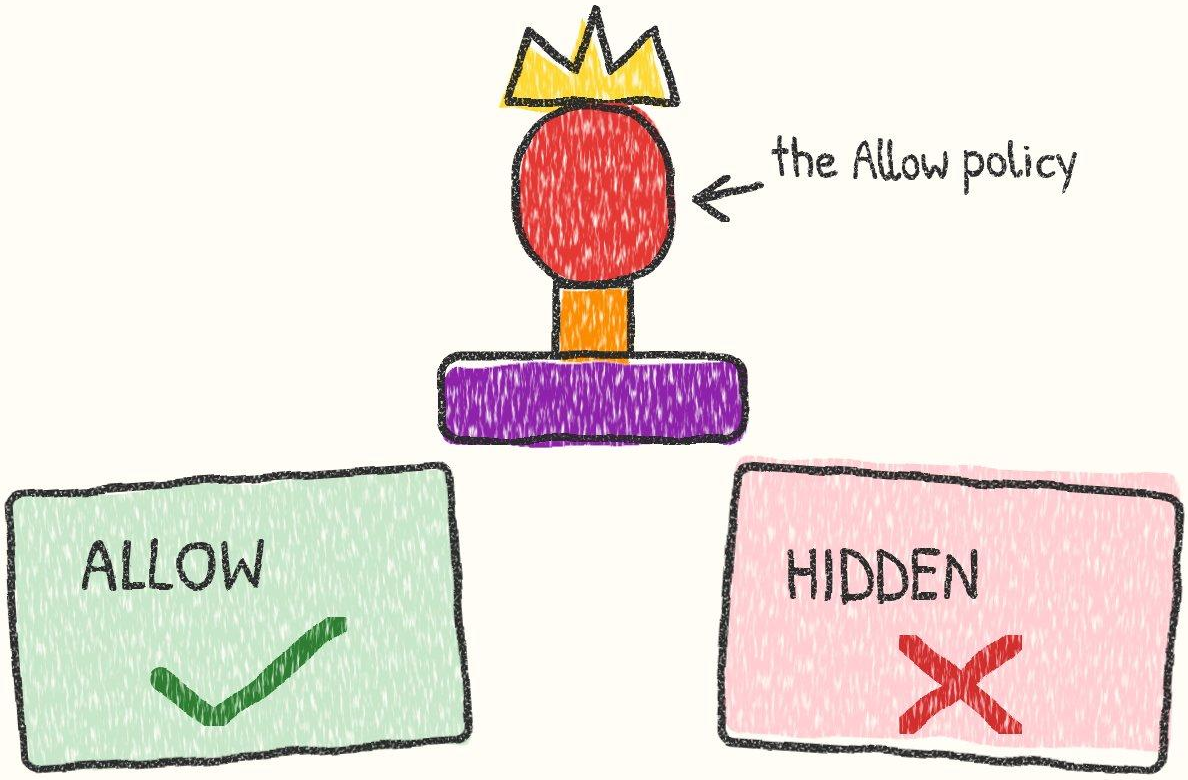


People log in through Cloudflare Access with the company sign-in they already have. Apps that are not browsers get a polite 401 and do the OAuth dance. Robots talking to robots bring a service token.

Color me: badge shiny, robot proud of its key

Grown-up version: Cloudflare MCP server portals docs, authentication

You Only See Your Toys

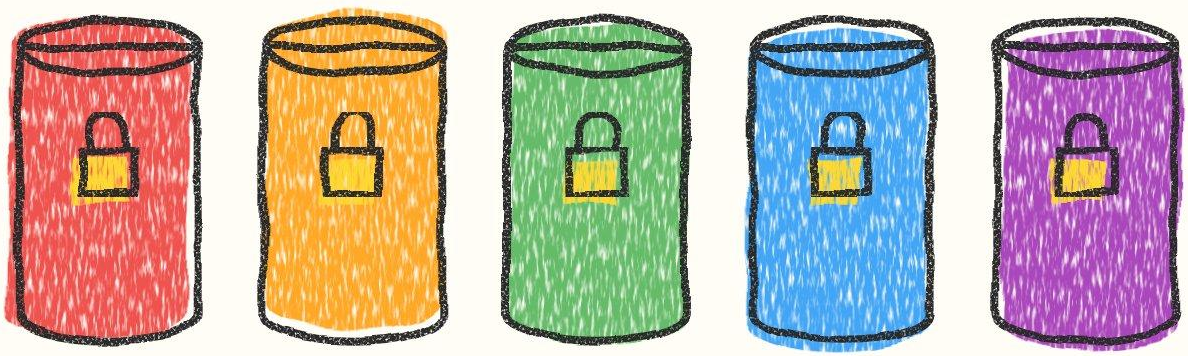


Every MCP server has its own Access policies. If you match the Allow policy, that server shows up in your portal. If you don't, it is not there at all. Not even a little.

Color me: ALLOW = green, HIDDEN = very, very quiet

Grown-up version: Cloudflare MCP server portals docs, Access policies

Every Server Has Its Own Lock



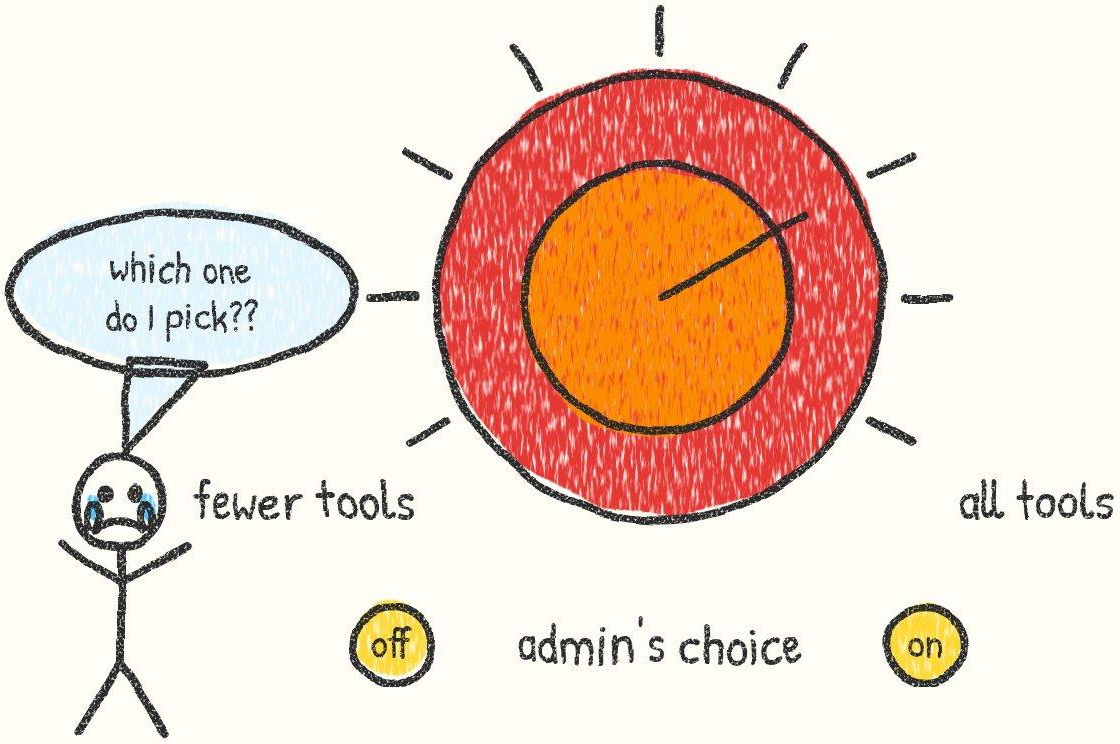
portal key + one key per server

You log in to the portal once. Then each server that wants its own OAuth sign-in asks you, just for itself. One big key for the portal, plus a little key per server.

Color me: one color per lock, no sharing keys

Grown-up version: Cloudflare MCP server portals docs, upstream OAuth

Turn Off The Scary Tools

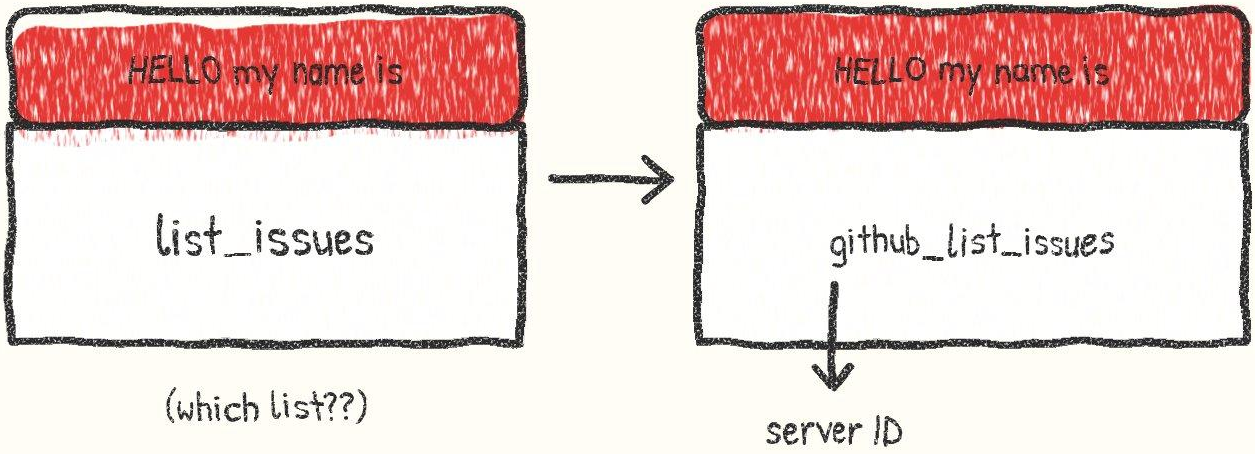


By default, every tool from a server shows up. Admins can switch single tools off, or flip to an allowlist and name only the tools people should get.

Color me: the knob a calm, responsible orange

Grown-up version: Cloudflare MCP server portals docs, tools and prompts

Tools Get Name Tags



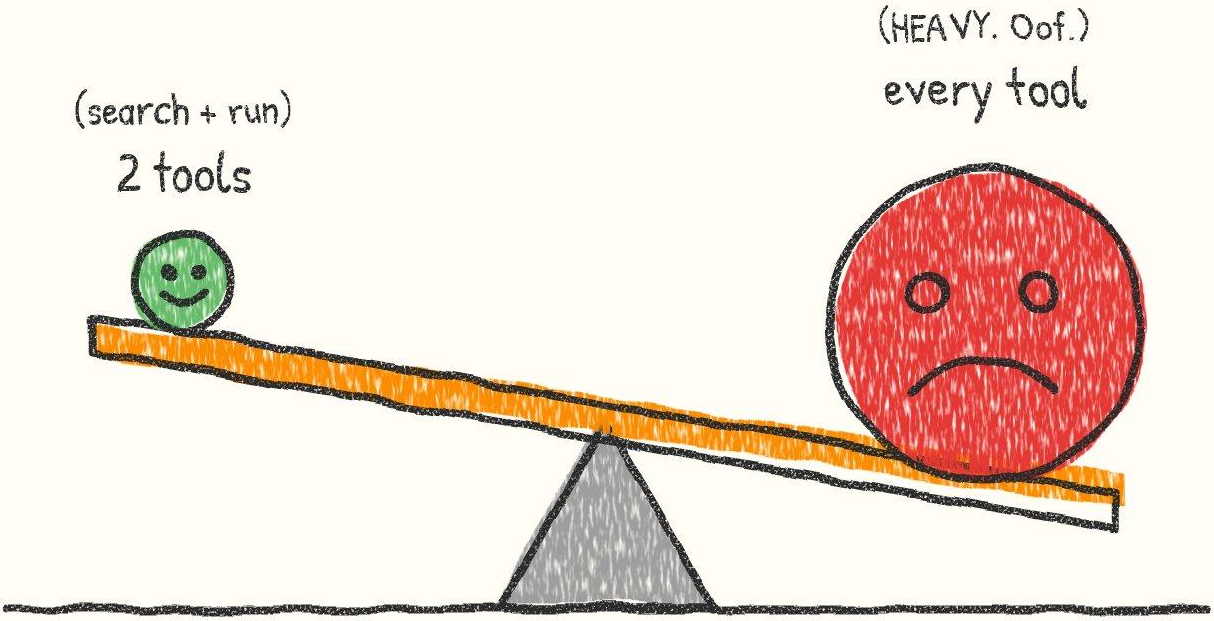
no underscores in server IDs, please

Each tool gets its server's name glued on the front, like github_list_issues, so two servers can both have a list tool. That is why server IDs can't have underscores. Admins can also give tools nicknames.

Color me: HELLO in red, like a real sticker

Grown-up version: Cloudflare MCP server portals docs, tool names and aliases

All The Tools In Two Tools

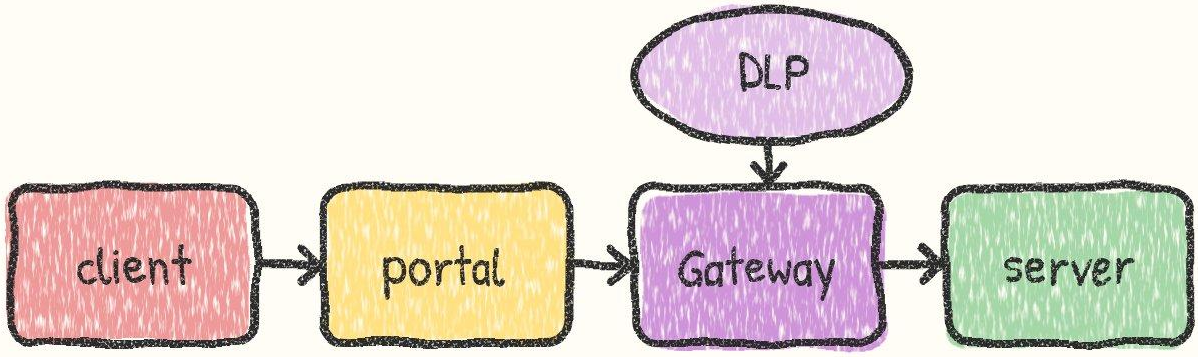


Code Mode squishes all the upstream tools into just two: one to search and one to run code. The AI writes a little JavaScript, and it runs in its own sealed-off Dynamic Worker. Admins decide if it is off, optional, on by default, or required.

Color me: the heavy ball the heaviest color

Grown-up version: Cloudflare MCP server portals docs, Code Mode

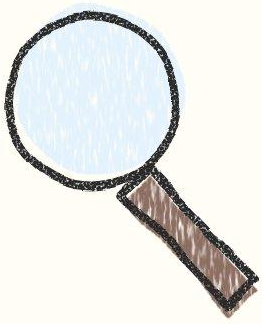
Take The Gateway Road



HTTP logs: yes

DLP peeking: yes

turned on: by you



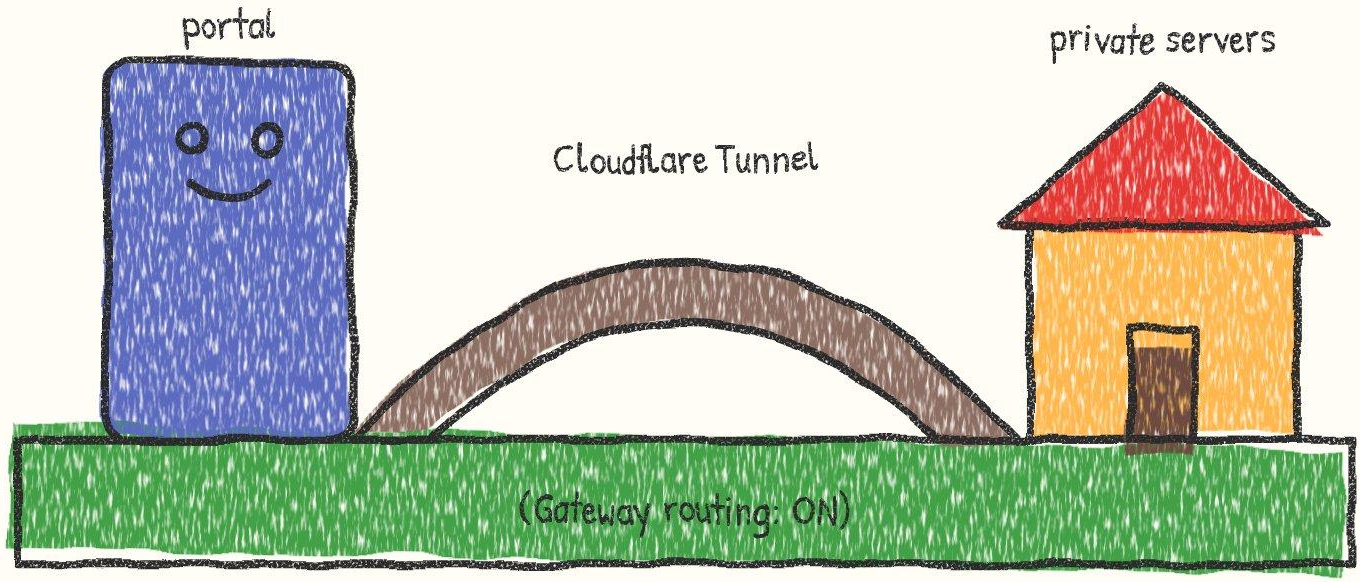
(optional!)

Gateway routing is optional. Turn it on for a server or a whole portal, and the traffic rides through Cloudflare Gateway, which writes HTTP logs and checks for data that should not leave the building.

Color me: bubbles in the DLP cloud

Grown-up version: Cloudflare MCP server portals docs, Gateway routing

The Secret Clubhouse

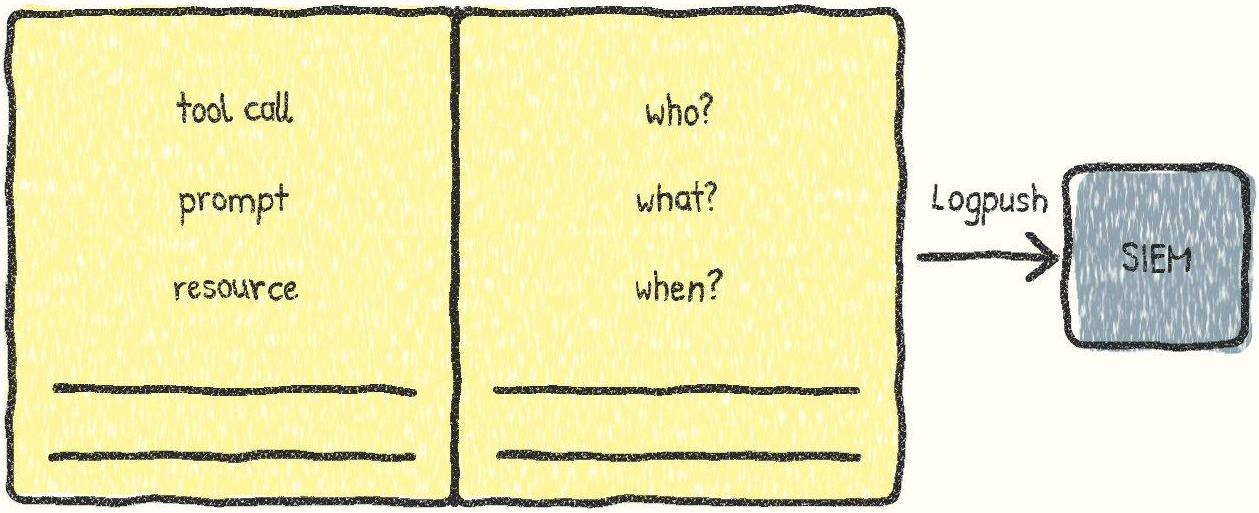


Some MCP servers live only on your private network. They connect through Cloudflare Tunnel, and Gateway routing has to be on. The OAuth sign-in doors still need to be reachable from the internet.

Color me: tunnel brown, clubhouse your favorite secret color

Grown-up version: Cloudflare MCP server portals docs, private MCP servers

Everything Goes In The Diary



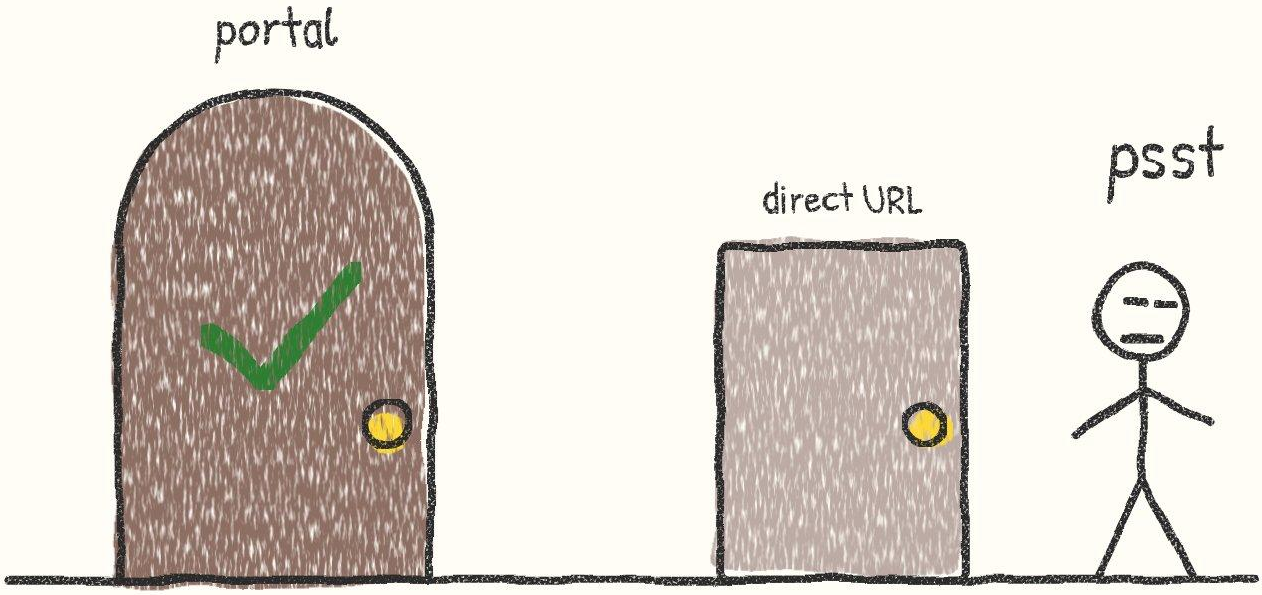
Dear diary: someone ran a tool.

Access writes down every request made through portal tools: the tools, the prompts, and the resources. Logpush can ship the whole diary to storage or to your SIEM, so the security grown-ups can read it later.

Color me: diary yellow, secrets in invisible ink

Grown-up version: Cloudflare MCP server portals docs, logs

The Back Door Is Still There



fix: make Access the server's OAuth provider

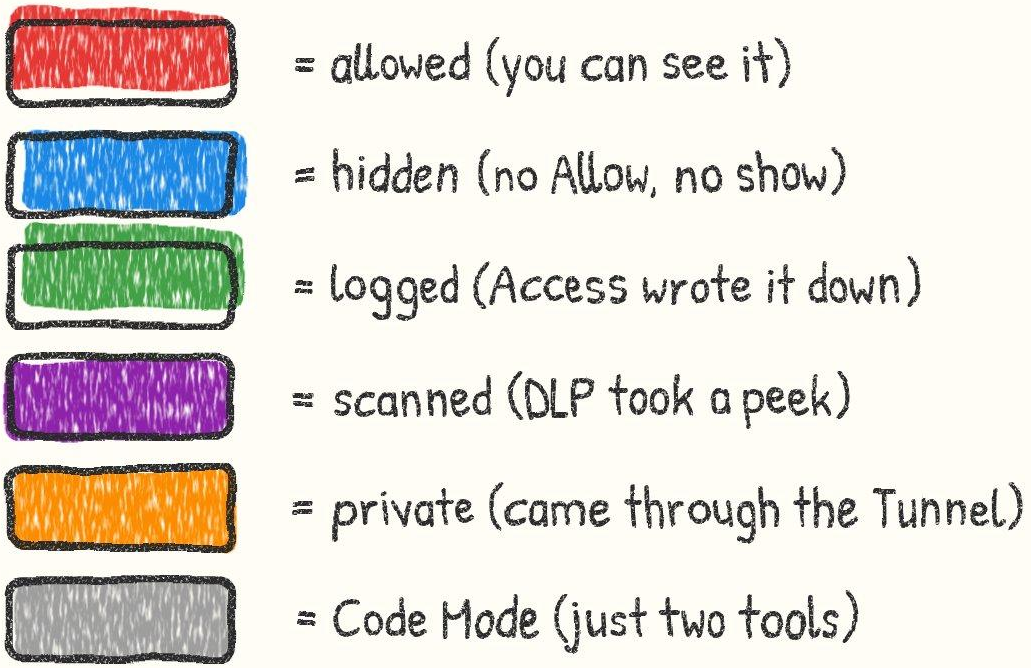
Careful! Someone the portal hides a server from can still knock on that server's own direct URL. To lock that door too, make Access the server's OAuth provider.

Color me: the sneaky one in sneaky green

Grown-up version: Cloudflare MCP server portals docs, the caution box

last page

Crayon Key & Sign-Off



= allowed (you can see it)

= hidden (no Allow, no show)

= logged (Access wrote it down)

= scanned (DLP took a peek)

= private (came through the Tunnel)

= Code Mode (just two tools)

Pop quiz: what does a portal URL end with?

_____

Signed, your friendly Zero Trust admin: _____

gold star (status: Ready)

THE END. Adapted from Cloudflare's MCP server portals docs and blog (CC BY 4.0). Not made or endorsed by Cloudflare.